

VAYUSETU: Technical Documentation

Installation Manual, API Reference, & Operation Guidelines

1. Installation Guide

To run VAYUSETU locally, follow these steps:

- Clone the repository to your workspace: `d:\BUNNY\PROJECTS\ISRO`
- Install backend dependencies: `pip install -r requirements.txt`
- Start the backend API gateway: `python backend/app/main.py`
- Navigate to the frontend folder: `cd frontend` and install modules: `npm install`
- Launch the local Vite web server: `npm run dev`

2. Main API Endpoints Reference

- **GET /api/v1/climate/live-state:** Fetches dynamic district climate vectors, runs Kalman corrections, and updates DB records.
- **GET /api/v1/prediction/forecast:** Serves multi-model ensemble forecast arrays with upper/lower uncertainty bounds.
- **GET /api/v1/simulation/runoff:** Runs what-if policy simulations, returning before/after metric deltas.
- **POST /api/v1/prediction/retrain:** Re-calibrates ConvLSTM and Transformer models against new observations and clears drift alarms.

3. Troubleshooting

Issue: The map repeats infinitely when zooming out.

Resolution: Ensure the Leaflet map bounds are clamped to `[5.0, 60.0]` and `[40.0, 100.0]`, and the tile layer has `noWrap: true` enabled.

Issue: Multi-model ensemble stuck at loading.

Resolution: Check if the backend port 8000 is accessible. The frontend has a built-in fallback state that automatically populates standard grids if the backend connection fails.